

Dimitri Chrysafis

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Education

University of Wisconsin–Madison
Bachelor of Science in Computer Science

Graduating May 2027
GPA: 4.0

Experience

SimpleX Chat

Open Source Contributor

- Contributed to SimpleX Chat, an open-source private messaging platform, by improving multi-device sync logic in the client messaging pipeline
- Implemented sync-state reconciliation and connection lifecycle updates to reduce duplicate delivery edge cases across linked devices

Projects

Interactive 3D Fluid Simulator Using Material Point Method [\[Demo\]](#) [\[GitHub\]](#)

- Built browser-based physics engine implementing Material Point Method (MLS-MPM) with WebGPU compute shaders, simulating **400,000+ particles** in real-time with interactive camera controls and dynamic boundary animations
- Optimized particle-to-grid transfers through GPU parallelization and fixed-point arithmetic, achieving smooth performance for dam break scenarios and real-time particle injection

High-Performance Web-Based Fractal Renderer [\[Blog\]](#)

- Engineered real-time fractal visualization for Mandelbrot, Newton's, and Kleinian limit sets using GPU-accelerated WebGL fragment shaders optimized for deep zoom
- Achieved up to **60×** speedup through adaptive sampling, tile-based rendering, and optimized complex arithmetic

Volumetric Sphere Packing for 3D Meshes [\[GitHub\]](#) [\[Blog\]](#)

- Built mesh-based sphere packing system that fills arbitrary 3D models with non-overlapping tangent spheres using ray-mesh containment tests and nearest-surface distance queries
- Accelerated candidate search and clearance updates with PyTorch tensor operations, Open3D mesh processing, and live visualization for iterative packing of complex OBJ meshes

Tennis Match Prediction System

- Built ensemble classifier using CatBoost on **92,000 ATP matches (1982-2024)** obtained from publicly available historical datasets, engineering features including player ELO ratings, surface-specific performance metrics, head-to-head statistics, recent form indicators, and fatigue scores from tournament schedules to achieve **83.7% prediction accuracy**, outperforming baseline models by **12%**

Fourier Drawing Machine [\[GitHub\]](#) [\[Blog\]](#)

- Implemented a full Discrete Fourier Transform pipeline in Python that converts sampled drawing points into complex coefficients and reconstructs curves with rotating epicycles
- Built an animation renderer that generates frame-by-frame trajectory output with PIL/NumPy and exports smooth drawing visualizations from custom or image-derived point sets